

AI CRM Lead Automation

Python 3.11+ local reference application | Standard-library runtime | Optional OpenAI Responses API | MIT licensed

DELIVERY	ANALYSIS	QUALITY	BOUNDARY
Local dashboard + API	Offline or optional OpenAI	39 automated tests	Reference app, not SaaS

Architecture

INTAKE	ANALYSIS	PERSISTENCE	DELIVERY
Threaded local HTTP server, strict JSON media type, request/message bounds, normalized object schema.	Deterministic default or strict Structured Outputs; local enums and text limits always enforced.	Process lock, duplicate check, atomic JSON replacement, raw message opt-in only.	Escaped dashboard, JSON endpoints, metrics, event log, formula-safe CSV.

HTTP surface

METHOD	ROUTE	PURPOSE
GET	/ and /api/health	Dashboard and mode-aware health check
POST	/api/intake	Validate, classify, score, and persist one lead
GET	/api/leads and /api/logs	Structured CRM records, metrics, and audit events
GET	/export/leads.csv	Formula-neutralized spreadsheet handoff

Security and privacy controls

- Loopback-only host validation	- HTML and CSS output constraints
- Bounded request body and message	- CSV formula neutralization
- Safe client-facing errors	- No secrets or lead text in logs
- Security and no-cache headers	- Raw inquiry retention off by default

Verification

39 isolated tests cover deterministic analysis, contact extraction, strict provider output, retries and refusals, corruption handling, duplicate events, atomic/concurrent writes, CSV safety, HTML safety, HTTP limits, routes, methods, errors, and headers. CI runs on Python 3.11-3.13 and rebuilds both PDFs.

Run locally

```
python3 -m unittest discover -s tests -v      python3 src/seed_demo.py      python3 src/app.py
```

Deployment boundary: single-machine reference implementation. Production use requires authentication, TLS, rate limits, managed persistence, monitoring, backups, and formal data lifecycle controls.